

Python Rule-Based Conversational Chatbot

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1. AIM

To develop a simple rule-based conversational chatbot in Python that continuously interacts with the user, recognizes different messages, provides appropriate responses, and exits when the user enters **bye** or **exit**.

2. OBJECTIVES

- Accept the user's name before starting the conversation.
- Use a continuous **while True** loop for interaction.
- Clean user input using **.lower().strip()**.
- Recognize greetings such as **hello** and **hi**.
- Respond to topics such as Python, studies, college, and careers.
- Use **if / elif / else** for decision making.
- Use the **in** operator for flexible inputs.
- Use **break** to stop the chatbot.

3. ALGORITHM / WORKING

1. Start the program.
2. Display a welcome message.
3. Ask the user for their name.
4. Enter the continuous conversation loop.
5. Read and clean the user's message.
6. Check for bye or exit.
7. Check greetings and supported topics.
8. If no condition matches, display an unknown-message response.
9. Repeat until bye or exit is entered.
10. Stop the program.

4. PYTHON CODE

```
print("Welcome! I am your study chatbot.")

name = input("What is your name? ")

print("Nice to meet you,", name)
print("You can ask me about college, Python, or studies.")
print("Type bye or exit to stop.")

while True:
```

```
message = input(name + ": ").lower().strip()

if "bye" in message or "exit" in message:
    print("Bot: Bye", name, "! Good luck with your studies.")
    break

elif "hello" in message or "hi" in message:
    print("Bot: Hi", name, "! How can I help you?")

elif "college" in message:
    print("Bot: College is a great place to learn new skills and meet people.")

elif "study" in message or "studies" in message:
    print("Bot: Try studying for a short time every day and practice what you learn.")

elif "python" in message:
    print("Bot: Python is beginner-friendly and useful for many applications.")

elif "job" in message or "career" in message:
    print("Bot: Keep improving your coding and communication skills for your career.")

else:
    print("Bot: I am still learning. Try asking me about Python, studies, college, or jobs.")
```

5. RESULT / TEST OUTPUT

The chatbot was executed successfully and tested with greetings, uppercase input, flexible greetings, different topics, an unknown message, and the exit command.

```
Welcome! I am your study chatbot.  
What is your name? aar thi  
Nice to meet you, aar thi  
You can ask me about college, Python, or studies.  
Type bye or exit to stop.  
  
aar thi: hello  
Bot: Hi aar thi ! How can I help you?  
  
aar thi: HELLO  
Bot: Hi aar thi ! How can I help you?  
  
aar thi: hey hello  
Bot: Hi aar thi ! How can I help you?  
  
aar thi: what is python  
Bot: Python is beginner-friendly and useful for many applications.  
  
aar thi: I want to study  
Bot: Try studying for a short time every day and practice what you learn.  
  
aar thi: tell me about college  
Bot: College is a great place to learn new skills and meet people.  
  
aar thi: I want a job  
Bot: Keep improving your coding and communication skills for your career.  
  
aar thi: I like cricket  
Bot: I am still learning. Try asking me about Python, studies, college, or jobs.  
  
aar thi: bye  
Bot: Bye aar thi ! Good luck with your studies.  
  
=== Code Execution Successful ===
```

6. CONCLUSION

The rule-based Python chatbot was successfully developed. It accepts user input, processes messages using string methods and conditional statements, responds to multiple topics, handles unknown messages, and terminates using the break statement.

7. EXECUTION SCREENSHOT

main.py

Visualize

Run

Output

```
1 print("Welcome! I am your study chatbot.")
2
3 name = input("What is your name? ")
4
5 print("Nice to meet you.", name)
6 print("You can ask me about college, Python, or studies.")
7 print("Type bye or exit to stop.")
8
9 while True:
10     message = input(name + ": ").lower().strip()
11
12     if "bye" in message or "exit" in message:
13         print("Bot: Bye", name, "! Good luck with your studies.")
14         break
15
16     elif "hello" in message or "hi" in message:
17         print("Bot: Hi", name, "! How can I help you?")
18
19     elif "college" in message:
20         print("Bot: College is a great place to learn new skills and meet people.")
21
22     elif "study" in message or "studies" in message:
23         print("Bot: Try studying for a short time every day and practice what you learn.")
24
25     elif "python" in message:
26         print("Bot: Python is beginner-friendly and useful for many applications.")
27
28     elif "job" in message or "career" in message:
29         print("Bot: Keep improving your coding and communication skills for your career.")
30
31     else:
32         print("Bot: I am still learning. Try asking me about Python, studies, college, or jobs.")
```

```
Welcome! I am your study chatbot.
What is your name? aarthi
Nice to meet you, aarthi
You can ask me about college, Python, or studies.
Type bye or exit to stop.
aarthi: hello
Bot: Hi aarthi ! How can I help you?
aarthi: HELLO
Bot: Hi aarthi ! How can I help you?
aarthi: hello bot
Bot: Hi aarthi ! How can I help you?
aarthi: hey hello
Bot: Hi aarthi ! How can I help you?
aarthi: what is python
Bot: Python is beginner-friendly and useful for many applications.
aarthi: I want to study
Bot: Try studying for a short time every day and practice what you learn.
aarthi: tell me about college
Bot: College is a great place to learn new skills and meet people.
aarthi: i want a job
Bot: Keep improving your coding and communication skills for your career.
aarthi: i like cricket
Bot: I am still learning. Try asking me about Python, studies, college, or jobs.
aarthi: bye
Bot: Bye aarthi ! Good luck with your studies.
```

=== Code Execution Successful ===